The AI Reasoning Ladder: What Machines Can Do — And Where They Still Fail

medium.com/data-science-collective/the-reasoning-ladder-what-ai-can-solve-today-and-what-remains-impossibly-hard-1c6791514bd9

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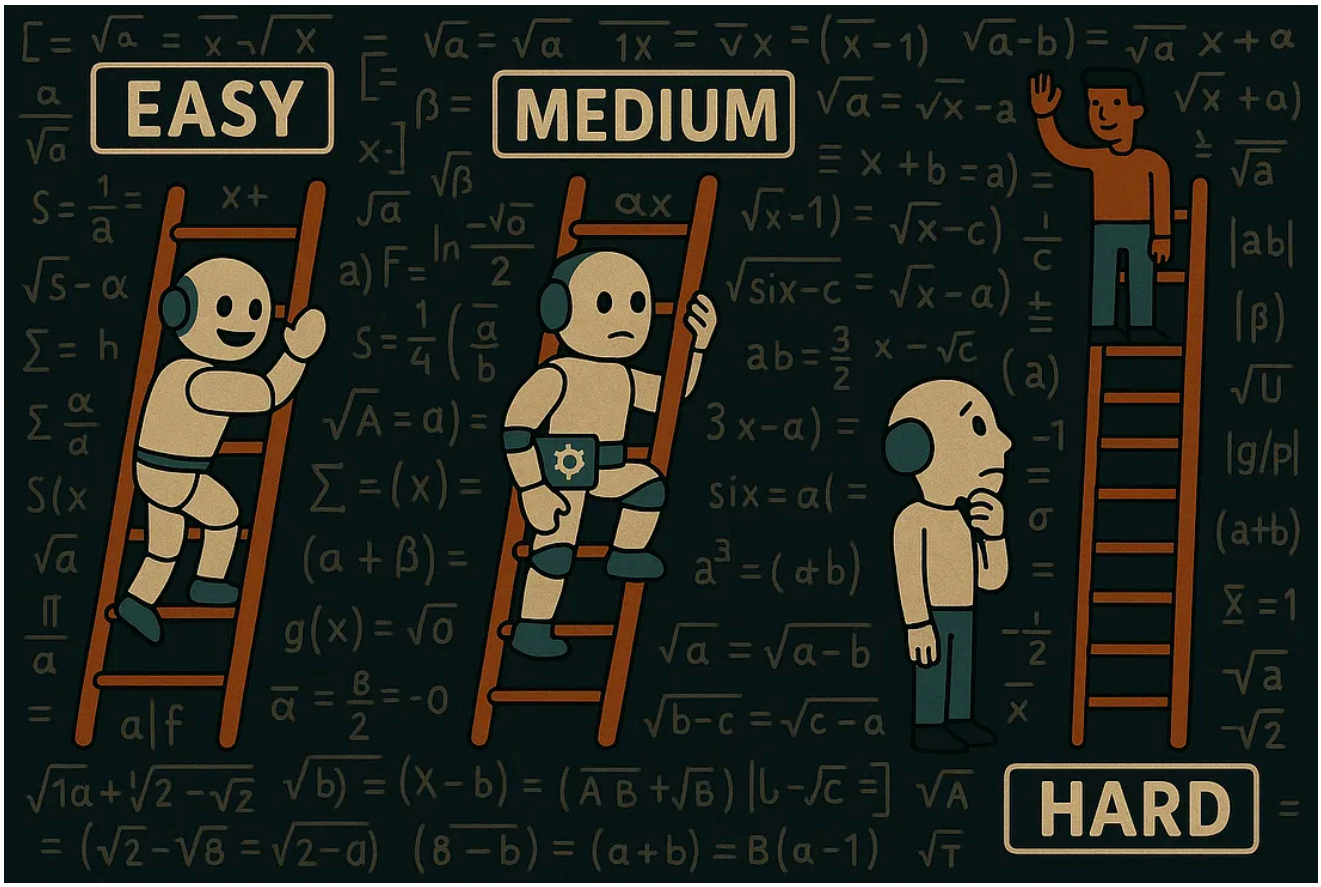

Imagine teaching a child to climb a tree. At first, they can only reach the lowest branches. With practice, they master climbing to the middle. Eventually, some reach the highest limbs, while others, despite all their training, can't reach the very top without a new technique or tool.

(Non-Member Link)

This is precisely what researchers discovered when they investigated how artificial intelligence handles increasingly difficult math problems.

In a fascinating new study, "*Climbing the Ladder of Reasoning: What LLMs Can — and Still Can't — Solve after SFT?*", scientists trained AI models on step-by-step math solutions and then tested them on progressively harder problems.

The results? With just a bit of training, these AI systems quickly learned to solve medium-difficulty problems that once stumped them. Yet as the problems grew harder, the AI hit clear limits, struggling with complex multi-step calculations and completely failing on problems requiring creative insights.

Just like our tree-climbing child, AI can only go so far with its current skills before it needs something fundamentally new.

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